Fiscal Policy

ECON 101 – Principles of Macroeconomics

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Chapter Outline

12.1 Learning Objective

Goal

Define fiscal policy and distinguish between automatic stabilizers and discretionary fiscal policy.

- **Fiscal policy:** changes in federal government taxes and spending intended to achieve macroeconomic objectives.
  - Provincial spending is important but primarily aimed at service delivery, not national macrostabilization.
- The federal government can **shift aggregate demand** by changing spending and tax decisions.
- **Canada 2020 example:** the federal government launched $314 billion in emergency spending (CERB, CEWS, CERS) — the largest fiscal intervention in Canadian history.

Automatic Stabilizers vs. Discretionary Fiscal Policy

Automatic Stabilizers

Spending and taxes that automatically change with the business cycle **without new legislation**.

- Employment Insurance (EI): rises in recessions as more workers qualify.
- Income tax: falls in recessions (lower incomes $\Rightarrow$ lower tax revenue).
- Social assistance: rises when unemployment increases.

Act fast, no legislative delay.

Discretionary Fiscal Policy

Intentional government actions requiring new legislation.

- 2009 Economic Action Plan: $40B infrastructure stimulus.
- 2020 CERB: $2,000/month to unemployed Canadians.
- 2024 GST holiday: temporary sales tax cut.
- 2025 income tax cut proposal.

Takes time; subject to political process.

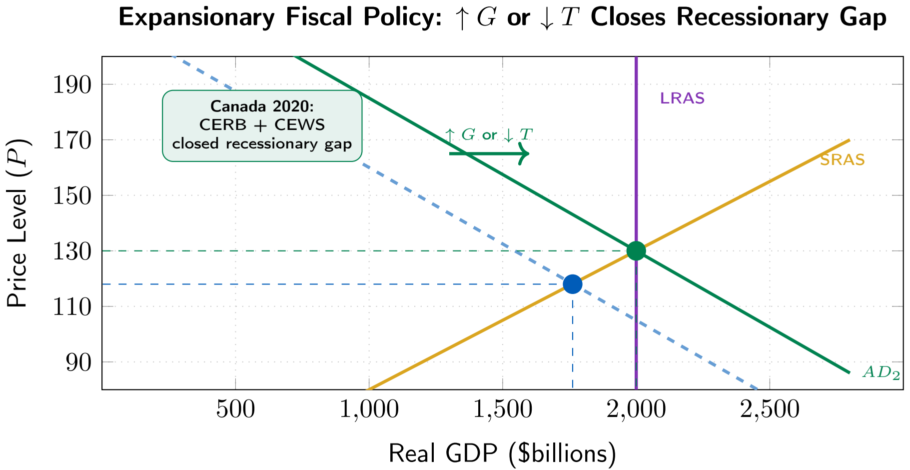
12.2 Learning Objective

Goal

Explain how fiscal policy affects aggregate demand and how it can stabilize the economy.

- **Two channels:**
  - Changes in government purchases (G): affects AD directly.
  - Changes in taxes (T): affects disposable income $\Rightarrow$ consumption $\Rightarrow$ AD indirectly.
- **Expansionary fiscal policy:** $\uparrow G$ or $\downarrow T \Rightarrow$ AD shifts right.
- **Contractionary fiscal policy:** $\downarrow G$ or $\uparrow T \Rightarrow$ AD shifts left.

Figure 12.1 – Expansionary Fiscal Policy (Closing a Recession)



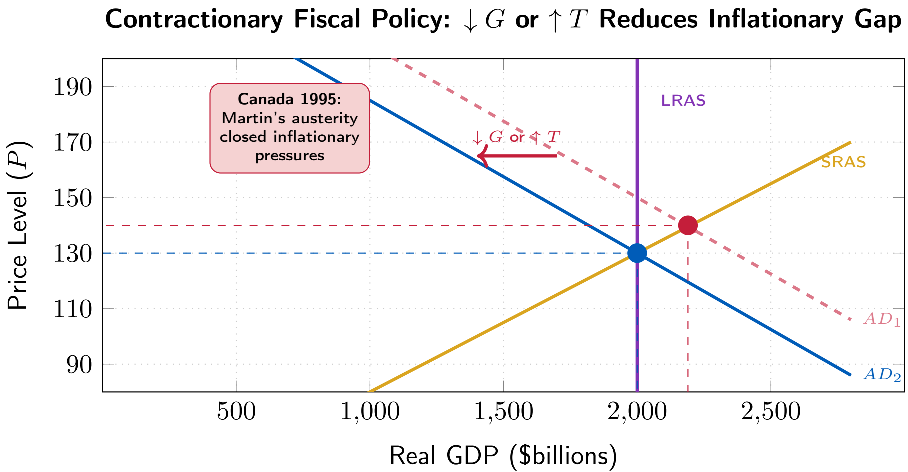

Contractionary Fiscal Policy: $\downarrow G$ or $\uparrow T$ Reduces Inflationary Gap
Canada 1995: Martin's austerity closed inflationary pressures
$\downarrow G$ or $\uparrow T$
LRAS
SRAS
AD_1
AD_2
190
170
150
130
110
90
500
1,000
1,500
2,000
2,500
Price Level (P)
Real GDP ($billions)

Table 12.1 – Countercyclical Fiscal Policy in Canada

Problem	Policy Type	Government Action	Canadian Example
Recession	Expansionary	$\uparrow G$ or $\downarrow T \Rightarrow$ AD shifts right	CERB/CEWS 2020 ($314B)
Rising inflation	Contractionary	$\downarrow G$ or $\uparrow T \Rightarrow$ AD shifts left	1995 Martin budget cuts

- **Note:** in Table 12.1 the original text references the U.S. Congress. The Canadian equivalent is Parliament — specifically the House of Commons, which must pass all spending bills and tax changes.
- Countercyclical policy aims to **dampen** the business cycle, not eliminate it.

Goal

Explain how the government purchases and tax multipliers work.

- A \$1 increase in government spending raises equilibrium GDP by more than \$1.
- Why?
  - Initial spending $\Rightarrow$ incomes rise $\Rightarrow$ consumption rises $\Rightarrow$ incomes rise further $\Rightarrow$ ...
  - This chain is the **multiplier effect**.
- **Government purchases multiplier:**

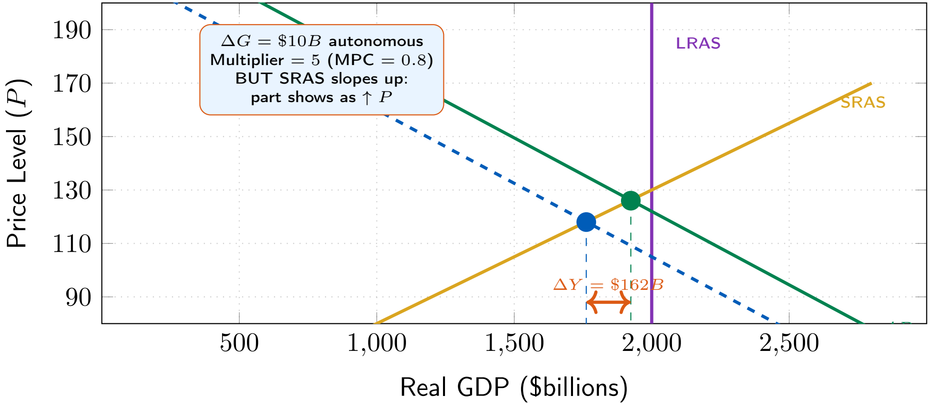
$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}.$$

- **Tax multiplier:**

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}}.$$

Figure 12.3 – The Multiplier Effect in AD-AS

Multiplier: \$10B Increase in G Raises Equilibrium GDP by \$50B (MPC $= 0.8$)



Government Purchases Multiplier vs. Tax Multiplier

Government Purchases Multiplier

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}$$

- \$1 of G **directly** adds \$1 to AD.
- Then the induced consumption chain kicks in.
- With MPC $= 0.8$: multiplier $= 5$.
- $\Delta G = \$10B \Rightarrow \Delta Y = \$50B$.

Tax Multiplier

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}}$$

- Tax cut raises disposable income.
- Households **save MPS** portion, spend MPC portion.
- First-round effect: \$MPC (not \$1).
- With MPC $= 0.8$: tax multiplier $= -4$.
- $\Delta T = -\$10B \Rightarrow \Delta Y = +\$40B$.

- The **government purchases multiplier is larger** (absolute value) than the tax multiplier: spending injection is fully spent, while a tax cut is partly saved.

12.5 Learning Objective

Goal

Discuss the difficulties that can arise in implementing fiscal policy.

- Fiscal policy is powerful in theory but faces practical limitations:
  1. **Timing problems** (legislative and implementation lags).
  2. **Crowding out** (higher government borrowing raises interest rates, reducing private spending).
  3. **Political constraints** (fiscal policy is a political process, not just an economic one).

Timing Problems

1. **Recognition lag:**
   - GDP data arrives 2–3 months after the quarter ends. By the time a recession is recognized, it may already be deep.
2. **Legislative delay:**
   - Parliament must debate and approve spending or tax changes.
   - **Canadian example (2009):** Conservative minority government's Economic Action Plan took months to implement after the recession had begun.
3. **Implementation delay:**
   - Infrastructure projects take months to years to actually begin (design, tender, construction).
   - **"Shovel-ready" projects:** a political priority in 2009 precisely because of this lag.
4. Because of these delays, fiscal stimulus may arrive after the recession has ended — adding fuel to the next boom.

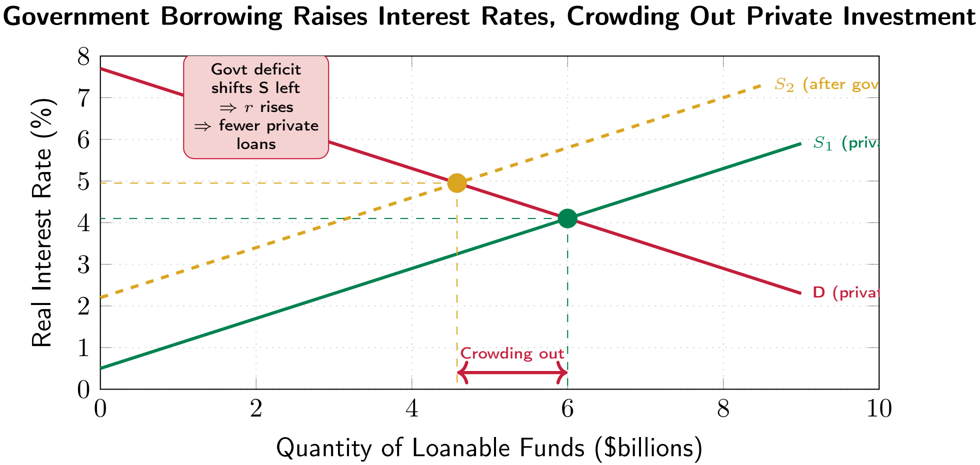
Crowding Out

Crowding Out

A decline in private expenditures (C, I, NX) that results from an increase in government purchases.

- **Mechanism:**
  - Government increases spending $\Rightarrow$ borrows more (issues more bonds).
  - Higher bond supply $\Rightarrow$ bond prices fall $\Rightarrow$ interest rates rise.
  - Higher interest rates $\Rightarrow$ lower private investment, lower consumer credit, stronger CAD (lower NX).
  - Private spending falls — partially offsetting the stimulus.
- **Canadian context:**
  - Canada's federal debt is ≈$1.3 trillion (2024).
  - Higher debt-to-GDP ratios may gradually raise borrowing costs.
  - But with BoC independence, crowding out is partially offset by monetary policy.

Figure 12.4 – Crowding Out in the Loanable Funds Market



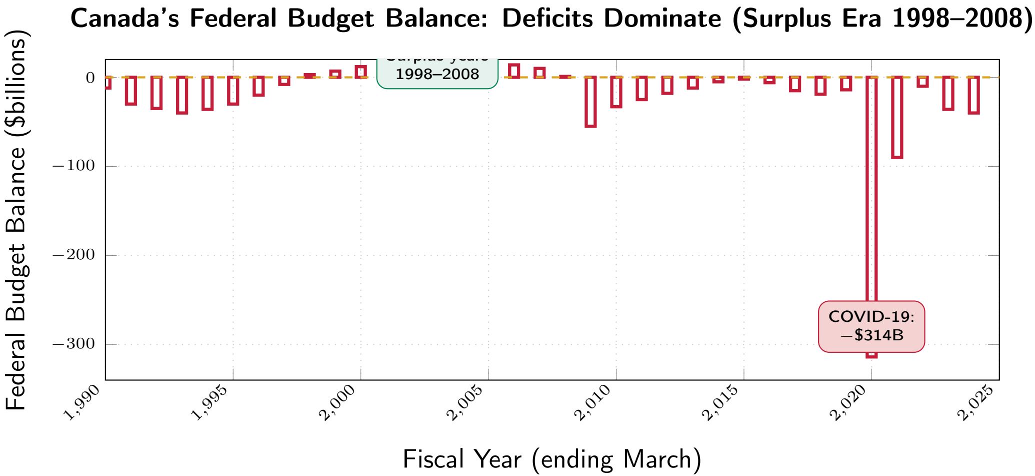
12.6 Learning Objective

Goal

Define budget deficit and federal government debt, and explain how the budget serves as an automatic stabilizer.

- **Budget deficit:** expenditures $>$ tax revenues.
- **Budget surplus:** expenditures $<$ tax revenues.
- **Federal government debt (national debt):** the total stock of outstanding Government of Canada bonds.
- **Canadian fiscal history:**
  - Deficits almost every year since 1970.
  - One stretch of surpluses: 1998–2008 (Chrétien/Martin era).
  - 2020–21: deficit of \$314 billion — by far the largest ever.
  - 2024–25: deficit $\approx$ \$40 billion.

Figure 12.5 – Canada's Federal Budget Balance (1990–2024)

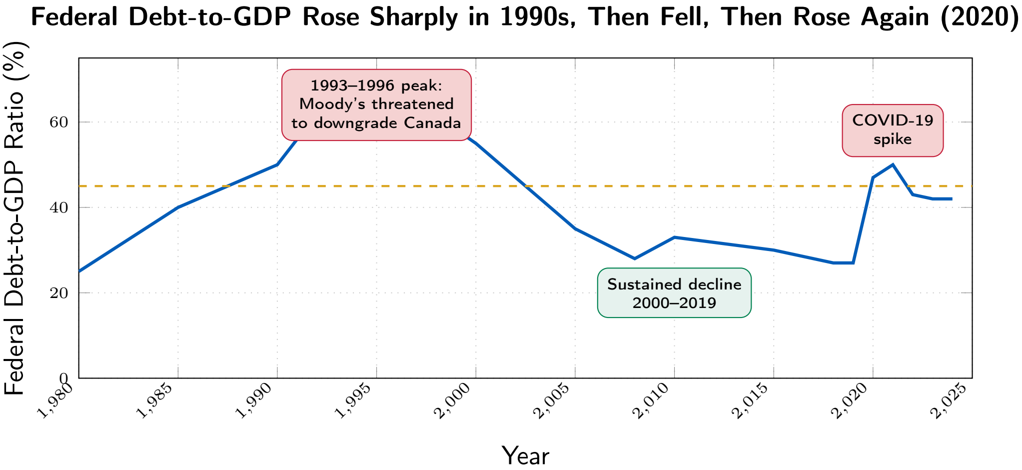


Source: Department of Finance Canada, Federal Budget. Positive = surplus; negative = deficit.

Automatic Stabilizers and the Business Cycle

- **In recessions** (automatic stabilizers kick in):
  - Tax revenues fall (lower incomes, profits, consumption).
  - EI payments, social assistance rise.
  - Result: budget deficit grows automatically, helping support AD.
- **In expansions**:
  - Tax revenues rise.
  - EI, welfare payments fall.
  - Result: budget moves toward surplus automatically, dampening inflation.
- **Canada's EI example (2020):**
  - EI and emergency benefits paid out $\approx \$100$ billion in 2020 alone.
  - This replaced lost income for millions of Canadians, supporting consumption.
  - Without this, the recession would have been **much deeper**.

Figure 12.6 – Federal Government Debt to GDP Ratio (Canada)



Source: Department of Finance Canada; IMF Fiscal Monitor. G7 average is approximate.

Is Government Debt a Problem?

- **Short-term view:** Canada's federal debt-to-GDP ratio ($\approx$42%) is lower than most G7 countries. Not an immediate crisis.
- **Long-term concerns:**
  - If debt grows faster than GDP, interest payments consume a larger share of the budget.
  - **Crowding out:** rising debt may gradually push up real interest rates, reducing private investment.
  - **Generational equity:** future taxpayers (today's children) inherit the debt.
- **When is debt less problematic?**
  - When it finances **productive investments**: infrastructure, education, R&D.
  - These raise future potential GDP, making the debt easier to service.
  - **Canada 2020:** emergency spending was necessary to prevent economic collapse, even though it was largely consumption-focused.

12.7 Learning Objective

Goal

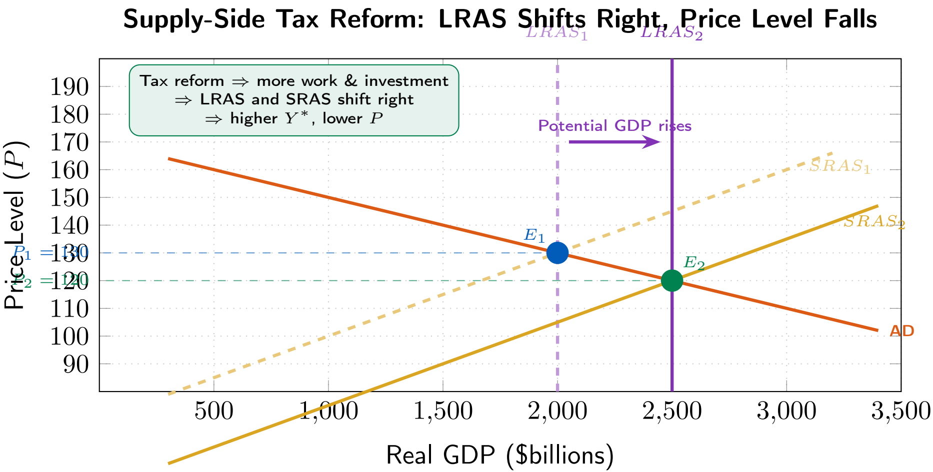
Discuss the effects of fiscal policy in the long run (supply-side effects).

- So far: fiscal policy as a short-run stabilization tool (shifts AD).
- Now: how fiscal policy affects **potential GDP** (shifts LRAS).
- **Supply-side economics:**
  - Fiscal policy actions aimed at increasing long-run aggregate supply.
  - Focus: taxes and regulations that affect incentives to work, save, invest, and innovate.

The Tax Wedge and Incentives

- Most taxes create a **wedge** between the pre-tax and post-tax return to an activity.
- **Examples:**
  - **Personal income tax:** if your marginal rate is 53% (federal + BC combined), each extra dollar earned yields only 47 cents after tax.
    - May reduce hours worked, entrepreneurship, or human capital investment.
  - **Corporate income tax:** Canada's combined rate ≈26% in 2024.
    - High rates may reduce investment in new machinery, R&D, and job creation.
  - **Capital gains tax:** affects saving and investment decisions.
    - **2024 Canada:** Trudeau government raised capital gains inclusion rate from 50% to 67% — controversial impact on investment incentives.

Figure 12.7 – Supply-Side Effects of a Tax Cut



Supply-side tax reform raises potential GDP from $2,000B to $2,500B while lowering the long-run price level.

Long-Run Effects of Tax Reform: What We Know

- Tax reform can increase potential GDP in the long run by:
  - Raising labour supply (more hours worked, more participation).
  - Increasing saving and investment (more capital formation).
  - Encouraging entrepreneurship and innovation.
- **Uncertainty:**
  - Magnitude of supply-side effects is **empirically contested**.
  - People may want to work more at lower marginal rates, but may face *constraints* (fixed schedules, childcare).
  - Tax cuts also create short-run demand effects (shifting AD right), which may dominate.
- **Canadian policy debate:**
  - 2024 capital gains inclusion rate increase: critics argued it would deter investment; supporters argued revenue would fund social priorities.
  - Active empirical research ongoing.

Appendix – The Government Purchases and Tax Multipliers

- Simple Canadian macroeconomic model (closed economy, fixed price level):

$$C = 1{,}000 + 0.8(Y - T), \quad T = 250,$$
$$I = 300, \quad G = 500.$$

- **Equilibrium:** $Y = C + I + G$:

$$Y = 1{,}000 + 0.8(Y - 250) + 300 + 500.$$

$$Y = 1{,}600 + 0.8Y - 200 = 1{,}400 + 0.8Y.$$

$$0.2Y = 1{,}400 \quad \Rightarrow \quad Y = \$7{,}000 \text{ billion}.$$

- **Government purchases multiplier:** $\frac{\Delta Y}{\Delta G} = \frac{1}{1-0.8} = 5.$
- $\Delta G = +\$10B \Rightarrow \Delta Y = +\$50B.$

The Tax Multiplier and the Balanced Budget Multiplier

- **Tax multiplier:**
$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1-\text{MPC}} = -\frac{0.8}{0.2} = -4.$$
- $\Delta T = -\$10B$ (tax cut) $\Rightarrow \Delta Y = +\$40B$.
- **Why smaller than G multiplier?** A \$10B tax cut gives \$10B to households, but they **save** $0.2 \times 10 = \$2B$, so only \$8B is initially spent.
- **Balanced budget multiplier:**
  - Increase G and T by the same amount (\$10B).
  - $\Delta Y = (+5) \times 10B + (-4) \times 10B = \$10B$.
  - Balanced budget multiplier $= \frac{\Delta Y}{\Delta G} = 1$.
  - Equal dollar increases in spending and taxes raise GDP by the same dollar.

Summary

- **Fiscal policy:** changes in federal spending (G) and taxes (T) to achieve macroeconomic goals.
- **Automatic stabilizers** (EI, progressive income tax) cushion downturns without legislation.
- **Expansionary** ($\uparrow G$ or $\downarrow T$) closes recessionary gaps; **contractionary** ($\downarrow G$ or $\uparrow T$) reduces inflationary pressure.
- Government purchases multiplier $= \frac{1}{1-\text{MPC}}$; tax multiplier $= -\frac{\text{MPC}}{1-\text{MPC}}$.
- Practical limits: **timing lags** (recognition, legislative, implementation) and **crowding out**.
- Canada's federal debt-to-GDP $\approx$42% (2024) — manageable but trending higher.
- **Supply-side** fiscal policy (tax reform) can shift LRAS right and raise potential GDP, but magnitudes are uncertain.
- **Balanced budget multiplier** $= 1$: equal increases in G and T raise GDP by the same dollar.
- **Canada 2020 case study:** \$314B in emergency spending was the largest ever fiscal expansion — essential to prevent depression-level output losses, at the cost of a large deficit and higher debt.

Thank you!

Questions?